



ADO.net

What is ADO.net?

ADO.NET (short for **ActiveX Data Objects** for .NET) is a set of data access components and libraries provided by Microsoft as part of the .NET Framework. It is used for **connecting to and manipulating data** from various data sources, such as databases, XML files, and web services, in applications developed using the .NET platform, including C#, VB.NET, and other .NET languages.

ADO.NET provides a way to interact with data in a disconnected and connected manner

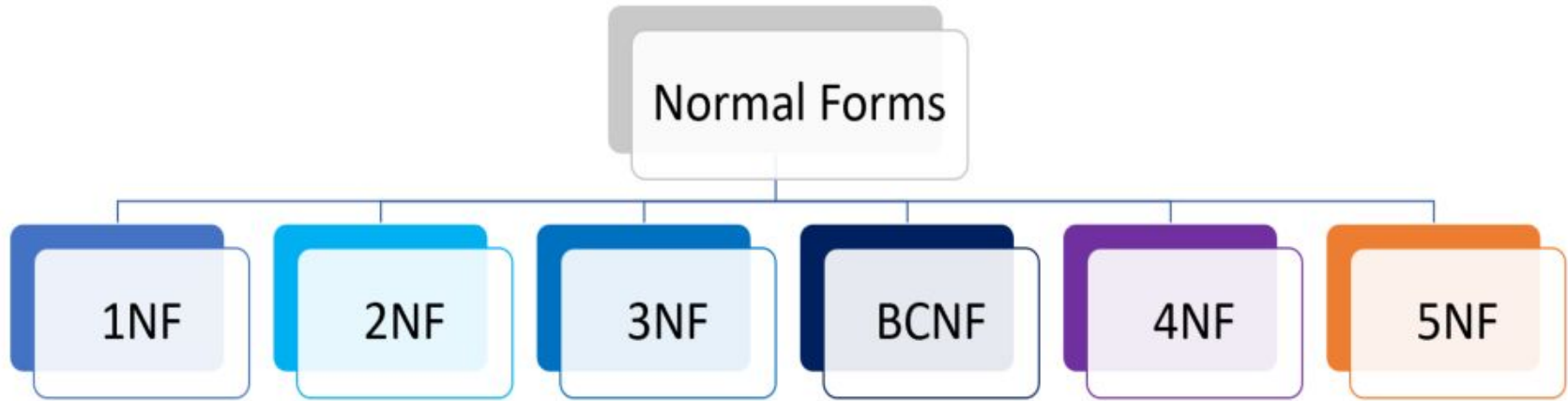
1. Connected Data Access:

- **Connection:** ADO.NET provides classes for establishing connections to various data sources, such as SQL Server, Oracle, MySQL, etc. The primary class for this purpose is SqlConnection for SQL Server databases.
- **Command:** You can create and execute SQL commands (queries, inserts, updates, and deletes) using classes like SqlCommand, OleDbCommand, and OracleCommand.
- **DataReader:** A DataReader allows you to retrieve data from a data source in a read-only and forward-only manner, which is particularly useful for querying large datasets.

2. Disconnected Data Access:

- **DataSet:** ADO.NET's DataSet is an in-memory representation of data retrieved from a data source. It can hold multiple DataTables, which are structured like database tables, along with relationships and constraints.
- **DataAdapter:** DataAdapters are used to populate a DataSet with data from a data source and to update changes made in the DataSet back to the data source. The SqlDataAdapter is commonly used with SQL Server databases.

Normal Form(NF)



First Normal Form(1NF)

- Single atomic value in each column
- Each row have unique identifier

1NF

Second Normal Form(2NF)

- Satisfy all 1NF conditions
- Partial dependencies must be removed from the table

2NF

Third Normal Form(3NF)

- Satisfy all conditions of 2NF
- Transitive dependency of non-key attributes on key column must be removed

Database	Application
Table	class
ستون	Property/Field
سطر	List<class>
کلید اصلی/کلید یکتا	Id

CSharp Data Types mapping SQL Server Data Type

<https://learn.microsoft.com/en-us/dotnet/framework/data/adonet/sql-server-data-type-mappings>

SQL Server	.NET
bigint	long
binary	byte[]
bit	bool
char	string
date	DateTime
datetime	DateTime
datetime2	DateTime
datetimeoffset	DateTimeOffset
decimal	decimal
filestream	byte[]
float	double
geography	Microsoft.SqlServer.Types.SqlGeography
geometry	Microsoft.SqlServer.Types.SqlGeometry
hierarchyid	Microsoft.SqlServer.Types.SqlHierarchyId
image	byte[]
int	int
money	decimal
nchar	string
ntext	string
numeric	decimal
nvarchar	string
real	Single
rowversion	byte[]
smalldatetime	DateTime
smallint	short
smallmoney	decimal
sql_variant	object
text	string
time	TimeSpan
timestamp	byte[]
tinyint	byte
uniqueidentifier	Guid
varbinary	byte[]
varchar	string
xml	string
else	object

Create Database

Create Table

```
CREATE TABLE table_name (  
    column1 datatype,  
    column2 datatype,  
    column3 datatype,  
    ....  
);
```

Create Table

SQL Insert into

```
INSERT INTO table_name (column1, column2, column3, ...)  
VALUES (value1, value2, value3, ...);
```

Update record

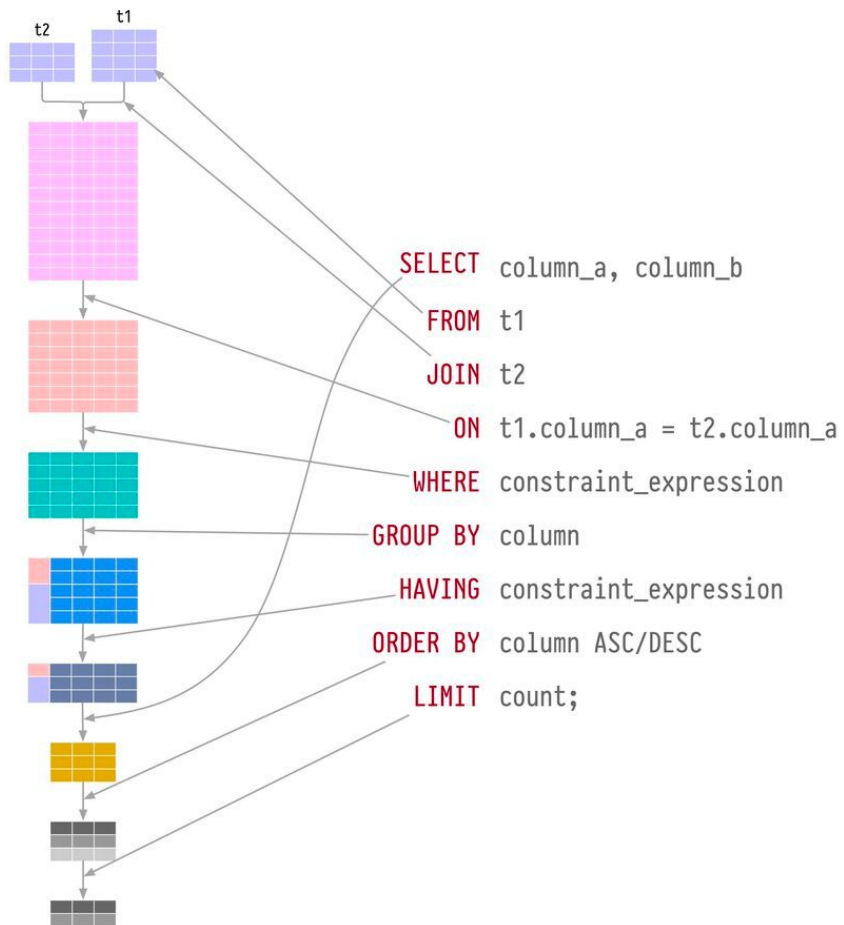
```
UPDATE table_name
```

```
SET column1 = value1, column2 = value2, ...
```

```
WHERE condition;
```

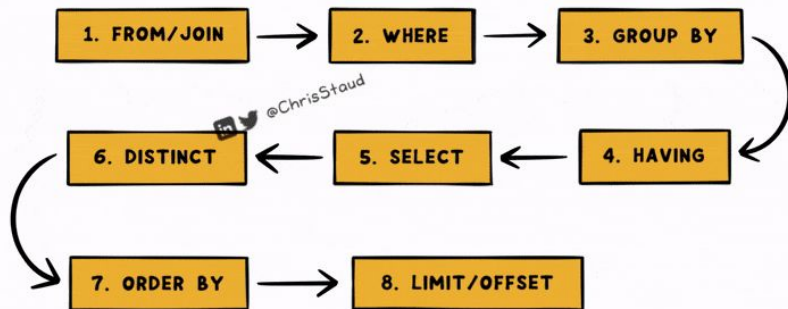
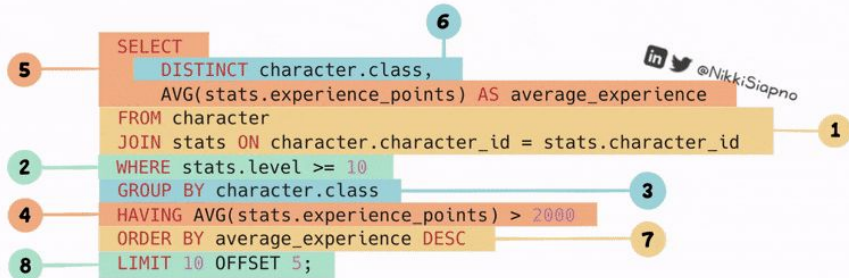
Delete

```
DELETE FROM table_name WHERE condition;
```



SQL Execution Order

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SQL Injection

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Explained

Username:

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```
SELECT * FROM users  
WHERE username='USER.INPUT';
```

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